

CE807 – Assignment

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Abstract

Sentiment analysis, also known as opinion mining, is a field of study that analyses people's opinions, sentiments, attitudes, and emotions expressed in written text (Liu, 2020). It is a natural language processing technique that aims to determine the polarity of a given text, whether it is positive, negative, or neutral. Sentiment analysis has a wide range of applications across various industries. For instance, businesses can use sentiment analysis to evaluate the success of their marketing campaigns by analysing the sentiment of online discussions and social media mentions (GeeksforGeeks, 2020). Positive sentiment indicates that the campaign is resonating with the target audience, while negative sentiment may indicate the need for adjustments (GeeksforGeeks, 2020). Another application of sentiment analysis is in customer feedback analysis. By analysing sentiment in customer reviews, social media posts, or conversations with service agents, businesses can gain valuable insights into how customers perceive their brand and make informed decisions to improve their products, services, or customer support strategies (Qualtrics, 2024). In this assignment, I performed text classification to analyse the sentiment of automotive reviews from an e-commerce dataset that means dealing with user feedback about vehicles or related products sold online. I utilized both unsupervised and discriminative approaches, including K-means clustering for unsupervised learning and Support Vector Machines (SVM) for discriminative model. Additionally, I explored other models such as topic modelling, Convolutional Neural Networks (CNN), but because it is computationally heavy, I opt to eliminated it, and Fully Connected Neural Networks (FCNN) to enhance the classification performance. These methods collectively aimed to accurately identify the sentiment expressed in the reviews.

Materials

- [Code](#)
- [Google drive folder](#)
- [Video presentation](#)

1 Task 1: Model Selection

1.1 Summary of 2 selected Models

The first model I selected was Support Vector Machine (SVM) which is a supervised and discriminative machine learning algorithm used for classification tasks. It is a powerful tool that can handle linearly separable as well as non-linearly separable data. The main objective of the SVM algorithm is to find the optimal hyperplane in an N-dimensional space that can separate the data points into different classes in the feature space (Sasidharan, 2021). The hyperplane is the decision boundary that separates the classes of data points. SVM is effective when the data is linearly separable or when a linear approximation is sufficient (Tabsharani, 2023). SVM can ignore outliers and find the best hyperplane that maximizes the margin between classes (Sasidharan, 2021). However, Choosing the appropriate kernel function and tuning the hyperparameters of an SVM can be challenging (anon, 2024). The second model I selected was K-means. It is an unsupervised machine learning algorithm used to partition a set of data points into K clusters. The algorithm assigns each data point to the cluster whose centroid (mean) is closest to it. The goal is to minimize the within-cluster sum of squares, also known as inertia. K-means is a simple and computationally efficient algorithm that scales well to large datasets. However, K-means is sensitive to the initial placement of centroids, which can lead to different cluster assignments (GeeksforGeeks, 2019). The algorithm may converge to a local minimum rather than the global minimum, resulting in suboptimal clustering (Dabbura, 2018).

1.2 Critical discussion and justification of model selection

I opt to choose k-means for unsupervised model because of several advantages, one of the key advantages of the k-means algorithm is that it doesn't require labelled data, making it straightforward to classify and group similar data points (Păvăloaia, 2024). Another advantage of k-means is its efficiency in handling large-scale datasets, making it

ideal for initial data exploration and reducing the complexity of data (Rifai, 2024). K-means is also straightforward to implement, with a simple algorithmic process that makes it accessible and easy to use for various data analysis tasks (Moreno, 2021). I chose k-means for its advantages, including ease of use and computational efficiency. To classify the data into positive and negative categories, I set the number of clusters, k , to 2. In terms of using supervised model, I chose SVM because of many reasons. The first reason is that Support Vector Machines (SVM) used for sentiment analysis of social media data due to its effectiveness in handling high-dimensional feature spaces typical in text data. Moreover, SVM provides clear decision boundaries, making it well-suited for distinguishing between positive and negative sentiments with high accuracy, especially when labelled training data is available since it is used for both cases (Bhumika M., 2016). SVM is known for its robust performance in text classification tasks, particularly when dealing with high-dimensional data such as word vectors. SVM ability to create optimal decision boundaries makes it ideal for accurately distinguishing between positive and negative sentiments in labelled movie review datasets (Tieu, 2020). Therefore, I chose SVM for its strong performance and accuracy in handling high-dimensional text data, making it ideal for sentiment analysis. In other words, it can handle the large feature spaces common in text data, like word vectors, making it well-suited for sentiment analysis, my task. It also creates clear and optimal decision boundaries, which helps in accurately distinguishing between different sentiment classes, such as positive and negative. SVM works well with various types of data, including text, and can be applied to both linear and non-linear classification problems using kernel functions. and SVM excels in scenarios where labelled data is available, leveraging this information to improve the accuracy of sentiment classification and I have the labels in my train data I selected then FCNN neural networks seeking more accuracy it was chosen for comparison with SVM.

2 Task 2: Design and implementation of classifiers

I received a dataset containing customer opinions about a purchased product. This dataset includes training, validation, and test data. Both the training and validation sets have two columns: one with text showing customer opinions and another with labels indicating whether the sentiment is positive or negative. For effective sentiment analysis, I

followed these steps to process and analyse the customer opinions on purchased products:

2.1 Text Cleaning:

- **Lowercasing:** Converted all text to lowercase so that words like "Happy" and "happy" are treated the same.
- **Removing Punctuation and Digits:** Eliminated punctuation marks and numbers, which don't contribute much to the sentiment analysis in my opinion.
- **Stop word Removal:** Removed common words like "the," "is," and "and" that don't have significant meaning in sentiment analysis.
- **Tokenization:** Split the text into individual words or tokens for more detailed analysis.
- **Stemming/Lemmatization:** Reduced words to their base forms, such as changing "running" to "run," to focus on the core meaning of the words.

I explore other method of cleaning the text like removing special formatting tokens and any markers removing multiple space with single space removing URLs and emails. However, this additional cleaning method has no effect. Therefore, I remain with the essential sufficient methods.

2.2 Vectorization:

TfidfVectorizer: Transformed the cleaned text into numerical features using TF-IDF (Term Frequency-Inverse Document Frequency), which reflects how important a word is in a document relative to a collection of documents (Smith, 2018). I chose to use it as it was used widely before and provide its effectiveness. And the numerical vectors produced by TF-IDF can be used with a wide range of machine learning algorithms, including SVM and k-means the model that I chose to implement as well as deep learning models I used for all chosen models to make comparison between them with the same vectorizer.

2.3 Modelling:

K-Means Clustering I Applied K-Means with $k=2$ to cluster the text data into two groups (positive and negative sentiments). K-Means helped identify patterns in the text without needing pre-labelled data for clustering. Therefore, I used it for unsupervised learning.

Support Vector Machines (SVM) I used various parameters for SVM to classify the text data into positive or negative sentiments. The parameter settings that yielded the highest accuracy were chosen to ensure the best performance. Techniques

were used like grid search with cross-validation to search for the best parameters I used low and high values for parameters to notice the difference and observe the selected value then I go around the selected value and try to be close to it to see if that would change the models' parameters and enhance the accuracy then I closed the best collection for the grid search.

These steps help ensure that my sentiment analysis is accurate by preparing and processing the text data effectively and choosing the most suitable machine learning models for classification. I tried using neural network model, Fully Connected Neural Networks (FCNN), as well as topic modelling techniques to improve performance. However, I didn't achieve better results than with the SVM. Given its strong performance and computational efficiency, I would recommend using SVM for this classification task than the previous mentioned models. For topic modelling, if simplicity is important, k-means would be the best choice. We have unbalanced data in both the training and validation sets. This means there are many more examples of positive sentiment than negative sentiment. As a result, the model learns to classify positive sentiment more effectively but struggles with negative sentiment due to the lack of sufficient examples. The distribution of positive and negative labels is detailed in Table 1.

Data set	Total	Class A: POSITIVE	Class B: NEGATIVE
Train	3681	3030	651
Valid	454	374	80
Test	409	NaN	NaN

Table 11 data statistics number of positive and negative label.

The results as shown in table 2 demonstrate that both SVM and k-means predict a higher number of positive sentiments compared to negative sentiments across all datasets. This reflects the imbalance in the training and validation data, where positive examples are more prevalent. K-means tends to predict more positive cases in the training set, which aligns with its ability to classify positive sentiments more effectively due to the imbalance. For the validation and test sets, SVM predictions still show a stronger inclination toward positive sentiment. K-means also predicts a higher number of positive sentiments but with a more balanced approach in the validation and test datasets compared to SVM. The slight overestimation of negative sentiment in the validation set suggests that k-means might be less sensitive to the imbalance compared to SVM or to make sure of our thinking we should go deeper and

look to the true negative, true positive, false negative and false positive. by examining that we can make our final decision because it could be the false negative and false positive more and that indicate a low accuracy even if the number of predicted negative more in our case. Overall, both methods exhibit a bias towards predicting positive sentiment, highlighting the challenge of working with unbalanced datasets. Adjustments, such as resampling or weighting, might be necessary to achieve a more balanced prediction performance.

Data set	Svm	k-means
Train positive	3178	3387
Train negative	503	294
Valid positive	417	374
Valid negative	37	80
Test positive	379	378
Test negative	30	31

Table 2 dataset detail after modelling

I obtain the models results and it is shown in table 3. From the table we can understand that SVM is outperform all previous models it is even better than neural network model. Moreover, the worst performance was achieved by k-means model.

model	F1 score
SVM	0.849
k-means	0.706
FCNN	0.842

Table 3 all models' performance

3 Task 3: Analysis and Discussion

In terms of performance, SVM slightly outperforms FCNN, with k-means trailing behind. The small differences between SVM, FCNN suggest that two are generally effective, with SVM having a marginal advantage in this specific dataset. However, FCNNs, being neural network models, are typically more resource-intensive than SVMs and k-means. They require more computational power and longer training times. In the table below table 4 I demonstrate different example were classified models in the test set

Text	SVM	k-means	FCNN
Test 0	N	P	P
Text96	P	N	p
Text 408	p	n	p
Text 196	n	p	p

Table 4 some interesting examples labels classified by the models.

When using K-means clustering to group review texts, similar reviews are often clustered together.

For example, if most reviews in one cluster are positive, the algorithm might classify any new review in that cluster as positive, even if it contains some negative comments. And this is happening in our case. For instance, Text 0” The only thing I would change on this mat is the height of the lip because it is something you have to deal with when putting heavy items in the truck. On the other hand, it definitely keeps spills contained. At only \$20, however, this mat is worth every penny. If you investigate the cost of similar items elsewhere, you know exactly what I mean. I would definitely buy this again.” generally has a positive tone, indicating the customer likes the product. However, it also mentions a minor issue that, if fixed, would improve the experience. Therefore, despite the overall positive sentiment, the review’s suggestion for improvement might indicate a negative sentiment when focusing on product enhancement. For more accurate sentiment classification, it might be better to use a different approach, like Support Vector Machines (SVM), which can more precisely identify and weigh the positive and negative aspects of the review. I mean, although I do not have the actual label, I think it should be negative and SVM made the right prediction while other models fail, and it might be because data imbalance. In the second example text 96 “great” and 409 “great product! great price Imo” the sentence was it is great, but k means classify this as negative what happened is K-means Clustered Without Sentiment Understanding. It grouped reviews based on how frequently words appear (without understanding their meaning), so that might end up with clusters where "great" is placed with other negative words. And because I used Term Frequency-Inverse Document Frequency (TF-IDF) "Great" might have a high TF-IDF score in reviews that discuss positive features frequently, but if most of the text where "great" is used comes from a negative context, it could still end up in a negative cluster. In the example 196 the sentence was “It won’t push over, to use on the side.” It is classified as positive by k-means because we convert this phrase into features presence of specific words. Key features so it takes "won’t push over" (suggests stability)"to use on the side" (implies versatility or additional use). So, it classified as positive. Why? Because of the Similarity to Positive Phrases The phrase "It won’t push over" is similar in sentiment to the positive phrases that talk about stability and reliability. If K-means clusters phrases with features suggesting stability or reliability together, this phrase will end up in a cluster with other positive phrases. Overall, some of the misclassified sentences are because they are ambiguous not easy to classify and some because of the preprocessing approach that was used, in my opinion.

3.1 Justification of Model’s performance

The performance difference between K-means and the other models (SVM, FCNN) can be attributed to several factors like algorithm nature because K-means is a clustering algorithm, not designed specifically for sentiment analysis. It groups similar data points but doesn't learn sentiment patterns explicitly, which can lead to below required level performance for sentiment classification tasks another factor might be feature representation it may use basic text features that don't capture sentiment nuances effectively, whereas SVM and FCNN can leverage advanced embeddings and learn complex sentiment patterns. However, this can not be the case in my assignment because I used the same feature representation for all models. The third factor might be supervised vs. unsupervised learning. In other word, SVM and FCNN are supervised models that learn from labelled data, enabling them to distinguish between positive and negative sentiments more accurately. In contrast, K-means is unsupervised and does not utilize sentiment labels during training and that could be the reason that k-means failed in classifying the negative sentiment. The last factor could be contextual understanding Advanced models like FCNNs can capture contextual information and sentiment nuances better than K-means, leading to improved performance in sentiment classification. However, they are computationally heavy that is why SVM is the best model.

3.2 Example and other Analysis

The best performance was achieved by The Support Vector Machine (SVM) the highest F1 score. This means it performed the best in balancing precision and recall, correctly identifying both positive and negative sentiments in the text. The second best was FCNN close behind. The Fully Connected Neural Network also performed well, though slightly lower than SVM. FCNN are effective but may not capture complex patterns as well as CNN (I tried to use it but it is heavy and require computational power that I can’t provide and it take so much time to run). The last model K-mean it achieved Lowest Performance K-means clustering had the lowest F1 score. This is expected because K-means is not designed for sentiment classification but rather for clustering similar data points together. It doesn’t learn sentiment patterns but just groups similar texts, leading to poorer classification performance.

4 Summary

In conclusion, I was given a text data where I had to classify people's opinions on products they bought, determining whether the reviews were positive (satisfied and happy) or negative (unsatisfied). This task is crucial for improving products based on customer feedback. I received training, validation, and test datasets. After cleaning and preparing the data, I used several machine learning and deep learning model for sentiment analysis. The Support Vector Machine (SVM) model performed the best, while the K-means clustering method had the worst performance. In data cleaning and preprocessing step, I involved in removing unnecessary information and converting text into a format suitable for analysis. In my assignment Supervised models performed better than unsupervised and they are generally performed better than unsupervised models, like K-means, especially in detecting negative sentiment. There are many other cleaning and preprocessing techniques, as well as different models, that could improve accuracy. More experimentation with these methods is needed to find the best approach. Overall, the supervised models showed better accuracy, indicating the importance of labelled data in training effective sentiment analysis systems. More testing and exploring different models and techniques will help enhance the accuracy of sentiment classification.

4.1 Lessons Learned

The first thing that I learned from this assignment is that learning and improving in this field takes time and experience. It's important to experiment with different approaches and learn from the results. Simple models can be very effective and computationally efficient. Sometimes, a model with moderate accuracy can be better than a complex one that requires more resources. Moreover, I learned that imbalanced data (where one class is much more common than another) can affect model performance. Understanding this helps in making better decisions about data handling and model training. Overall, this assignment taught me the importance of preprocessing text properly, choosing the right features, and selecting the appropriate model and evaluation metrics for effective sentiment analysis. In this assignment, I gained crucial insights into sentiment analysis:

- **Text Preprocessing:** I learned to clean text by removing punctuation, converting to lowercase, and handling special characters. Tokenization breaks text into manageable pieces, and vectorization (using TF-IDF) converts text into numerical data for models.
- **Feature Engineering:** I discovered how to select meaningful features and reduce dimensionality to enhance model efficiency.
- **Model Building and Evaluation:** I explored various models (like SVM and FCNN), trained them while avoiding overfitting and underfitting, and used metrics like accuracy, precision, recall, F1-score, and confusion matrices to evaluate performance effectively.

References

- anon. (2024). Machine Learning Algorithms Explained: Support Vector Machine. *Machine Learning Algorithms Explained: Support Vector Machine*.
- Bhumika M., V. B. (2016). Sentiment Analysis using Support Vector Machine based on Feature Selection and Semantic Analysis. *Sentiment Analysis using Support Vector Machine based on Feature Selection and Semantic Analysis*, 26–30.
- Dabbura, I. (2018). K-means clustering: Algorithm, applications, evaluation methods, and drawbacks. *K-means clustering: Algorithm, applications, evaluation methods, and drawbacks*.
- et, a. S. (2018). Learning about strategies and difficulties of emotion regulation increased emotional well-being and belonging in school among middle schoolers over six weeks. *Learning about strategies and difficulties of emotion regulation increased emotional well-being and belonging in school among middle schoolers over six weeks*.
- GeeksforGeeks. (2019). K Means Clustering. *GeeksforGeeks*.
- GeeksforGeeks. (2020). What is Sentiment Analysis? *GeeksforGeeks*.
- Liu, B. (2020). *Sentiment Analysis*. <https://www.cambridge.org/core/books/abs/sentiment-analysis/introduction/563742A639EEE9F5AB3F29CB2387E41C>: Cambridge University Press.
- Moreno, a. (2021). Understanding Customers' Transport Services with Topic Clustering and Sentiment Analysis. . *Understanding Customers' Transport Services with Topic Clustering and Sentiment Analysis*. .
- Păvăloaia, D. (2024). Clustering Algorithms in Sentiment Analysis Techniques in Social Media -A Rapid Literature Review. . *Clustering Algorithms in Sentiment Analysis Techniques in Social Media -A Rapid Literature Review*. .
- Qualtrics. (2024). What is sentiment analysis and how to use it? *Qualtrics*.
- Rifai, H. (2024). The Sentiment analysis of Movie reviews using the Transfer Learning approach MSc Research Project Data Analytics Tanmay Shrikant Paranjape Student ID: x20101015 School of Computing National College of Ireland.
- Sasidharan, A. (2021). Support Vector Machine (SVM) Algorithm. *Support Vector Machine (SVM) Algorithm*.
- Smith, j. (2018). Learning about strategies and difficulties of emotion regulation increased emotional well-being and belonging in school among middle schoolers over six weeks. *Learning about strategies and difficulties of emotion regulation increased emotional well-being and belonging in school among middle schoolers over six weeks*.
- Tabsharani, F. (2023). What is support vector machine (SVM)? *What is support vector machine (SVM)?*
- Tieu, B. (2020). bennytieu/ml_movie_reviews.

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